

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output label structure is appropriate for the deployment's per-task classification paradigm [Q84, A4], and headline acceptability binary (True/False) [Q148] and TC's category labels [Q153] map cleanly to editorial use. However, several output-ontology features misalign with deployment needs (HIGH priority): (a) LA Multi exposes formally-derived linguistic error categories (Verb, Adj, Syntax, Ortho) [Q31, Q151] grounded in the LOD/Spellchecker formal standard, while the deployment explicitly applies a flexible journalistic register [A3] — the Ortho category in particular may penalise legitimate journalistic variants; (b) the GPE→LOC merge [Q44] reduces granularity for cross-border journalism that needs to distinguish municipalities, regions, and countries [WEB-1, regional YAML]; (c) the ANIMALS topic label [Q48] encodes a non-newsroom assumption; (d) intent detection's xSID schema [Q154] is operationally irrelevant. The benchmark explicitly cautions against using performance as evidence of cultural/demographic coverage [Q121].

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)

Strengths

- Label-output (not MCQA) paradigm [Q84] mirrors the deployment's per-task classification with valid label sets, matching the deployment's per-task scoring pipeline [A4]
- HA binary True/False [Q148] directly maps to the deployment's headline-flagging logic

- TC label set covers standard international news categories aligned with the deployment's editorial taxonomy [Q153, A1]

ID	Evidence
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: HA [Q148] and TC [Q153] labels align with deployment needs. NER tags (LOC/PER/ORG/MISC/DATE) [Q152] are appropriate categories but the GPE→LOC merge [Q44] is regionally suboptimal. LA Multi categories [Q151] encode formal linguistic error types divergent from the deployment's journalistic register [A3].

OO-2: GPE has been merged into LOC [Q44], removing a category Luxembourgish cross-border journalism plausibly needs to distinguish (municipalities vs. countries vs. geographic regions), per regional YAML and [WEB-1]. No journalist-register acceptability category exists.

OO-3: The LA Ortho category [Q31, Q151] encodes Spellchecker.lu's formal ZLS standard [WEB-11], which represents non-deployment values (formal orthographic correctness) rather than the editorial register the deployment applies [A3]. The ANIMALS TC label [Q48] reflects an editorial assumption inherited from RTL's category structure rather than universally newsroom-relevant.

OO-4: Stakeholder-driven taxonomy redesign is warranted for LA Multi (to recognise journalistically-acceptable variants) and for TC (to evaluate whether ANIMALS is needed and whether POLITICS / EU coverage labels should be added) given HIGH OO priority weight.

OO-5: Documented taxonomy issues: (a) LA Ortho derived from formal-standard Spellchecker.lu [Q31] vs. flexible journalistic register [A3]; (b) GPE→LOC merge [Q44] vs. cross-border granularity needs; (c) ANIMALS label [Q48] vs. standard newsroom beats; (d) intent detection schema [Q154] irrelevant to deployment.

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)

Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Information Gaps

- Whether LA Ortho ground-truth labels would survive review by professional Luxembourgish editors applying journalistic register
- Whether the deploying outlet's NER schema requires the GPE/LOC distinction

Requires Expert Verification

- Editorial review of a sample of LA Ortho instances against journalistic-register standards
- Editorial confirmation of the operational TC taxonomy

Remediation

Gap 1: LA Ortho ground truth derived from Spellchecker.lu / formal ZLS standard [Q31, WEB-11] diverges from the deployment's flexible journalistic register [A3]

Recommendation: Either drop the Ortho subclass for production scoring or remap it against an editorial style guide; involve the deploying outlet's chief sub-editor in defining what counts as an unacceptable orthographic variant

Gap 2: GPE has been merged into LOC [Q44], reducing granularity for Luxembourgish cross-border journalism that needs to distinguish municipalities, regions, countries, and EU institutions [WEB-1]

Recommendation: Restore the GPE/LOC distinction in a deployment-specific NER schema and supplement training data with annotated EU institutional and Grande Région entity examples

ID	Evidence
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)